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COURSE: DSA0501 – Query Processing for Data Science

1.Case Study: Monthly Hospital Patient Visits

A hospital administrator wants to analyze the number of patients visiting the hospital each month during 2025.

Monthly Hospital Patient Visits

Jan 1200

Feb 1350

Mar 1500

Apr 1450

May 1700

Jun 1850

Questions

1. Which visualization is most suitable?

=>A **line chart** best for showing trends over months.

2. What trend can be observed?

=>Patient visits are **increasing overall**, with slight dip in April.

3. Identify the month with the highest patient count.

=>**June (1850 patients).**

4. Predict whether visits are increasing or decreasing.

=>Visits are **increasing**, so future months likely continue rising.

2.Case Study: Study Hours vs Exam Marks

A teacher wants to determine whether studying more hours improves exam performance.

Student	Study Hours	Marks
A	2	45
B	3	52
C	4	58
D	5	68
E	7	82

Questions

1. Which chart is appropriate?

=>A **scatter plot** — shows relationship between two variables.

2. Is there a positive or negative relationship?

=>A **positive relationship** — more study hours → higher marks.

3. Which student is an outlier, if any?

=>No major outlier; all points follow the upward trend.

4. Can marks be predicted from study hours?

=>Yes — since the relationship is strong and positive, marks can be **reasonably predicted** using study hours (linear trend).

3.Case Study: Employee Salary Distribution

A company wants to understand how salaries are distributed among employees.

Salary Data (₹ in Thousands)

25, 28, 30, 32, 35, 35, 36, 38, 40, 42, 45, 48, 50, 52, 55, 60, 65, 70

Questions

1. Which graph should be used?

=>A **histogram** — best for showing salary distribution.

2. In which salary range do most employees fall?

=>Most values fall between **35–50 thousand**.

3. Is the distribution symmetric or skewed?

=>Slightly **right-skewed** (more lower salaries, fewer high salaries).

4. Are there any salary ranges with very few employees.

=>Very few employees in **60–70 thousand** range.

4.Case Study: Favorite Programming Language

A survey was conducted among 200 students.

Language Number of Students

Python 80

Java 50

C++ 30

JavaScript
25

t

R 15

Questions

1. Which visualization is suitable?

=>A **bar chart** — best for comparing categories.

2. Which language is most popular?

=>**Python (80 students)**.

3. Which language is least preferred?

=>**R (15 students)**.

4. Compare Python and Java popularity.

=> Python (80) is **much more popular** than Java (50).

5. Case Study: Household Monthly Expenses

A family spends money in the following categories:

Category	Expense (%)
Rent	40
Food	25
Education	15
Transportation	10
Entertainment	10

Questions

1. Which chart best represents the data?
=>A **pie chart** — shows percentage share clearly.
2. Which category consumes the largest budget?
=>**Rent (40%)**.
3. What percentage is spent on Food and Education together?
=> $25\% + 15\% = 40\%$.
4. Which categories occupy the smallest share?
=>Transportation (10%) and Entertainment (10%) — both smallest.

6. Case Study: Student Test Scores

Scores obtained by students:

45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

Questions

1. Which plot should be used?
=>A **box plot** — best for showing spread, median, and outliers.
2. What is the median score?
=>There are 15 scores → median is the 8th value: **Median = 65**
3. Are there any outliers?
=>95 is **much higher** than the rest → **a high outlier**.

4. What is the spread of the data?

=> Scores range from **45 to 95**, showing a **wide spread** with most scores between **55–80**.